

STAT 212 Problem Set 6.

Due: Wednesday, April 30th at 11:59PM

Instructions: Collaboration with your classmates is encouraged. Please identify everyone you worked with at the beginning of your solution PDF (e.g. Collaborators: Alice, Bob). Your solutions should be *written* entirely by you, even if you collaborated to *solve* the problems. The first person to report each typo in this problem set (by emailing me and Somak) will receive 1 extra point; more serious mistakes will earn more points.

In all problems below, B_t denotes standard Brownian motion, and $\mathcal{F}_t$ the associated filtration.

1. For the martingale CLT from class, we assumed the total variance $V = \sum_{k=1}^{m(n)} \mathbb{E}[X_k^2 | \mathcal{F}_{n,k-1}]$ equals 1 almost surely. Show the martingale CLT from class continues to hold if V converges in probability to 1. (**Hint** given on the last page.)
2. Consider d -dimensional Brownian motion $\vec{B}_t$ for $d \geq 3$, started with $\|\vec{B}_0\| = 1$.

- (a) For fixed $\epsilon \in (0, 1)$, let τ_ϵ be the first time that $\|\vec{B}_t\| = \epsilon$. Prove that

$$X_{t \wedge \tau_\epsilon} = \|\vec{B}_{t \wedge \tau_\epsilon}\|^{2-d}$$

is a local martingale.

- (b) Deduce that $\vec{B}_t$ almost surely never hits the origin.
- (c) Prove that X_t is not a true martingale. (**Hint** given on the last page.)
- (d) Show that in fact $\vec{B}_t$ must wander off to infinity and eventually never come back to any given bounded region (i.e. $\vec{B}_t$ is transient).

Note: this problem was updated after the initial posting and broken into more parts, but no additional statements to be proved were added.

3. Here you will construct spherical Brownian motion in $\mathbb{R}^d$, for $d \geq 2$. (Another construction in the $d = 2$ case was also given in the previous homework.) For $\|x\| > 0$, let $P_x^\perp$ be the projection matrix onto $x^\perp$, given explicitly by $P_x^\perp = I_d - \frac{xx^\top}{\|x\|^2}$.

You may use below that existence and uniqueness for SDE solutions as in class continues to hold (with exactly the same proof) in higher dimensions, for equations of the form

$$dX_t = \sigma(X_t, t)dB_t + v(X_t, t)dt$$

with $\sigma : \mathbb{R}^d \times \mathbb{R}_+ \rightarrow \mathbb{R}^{d \times d}$ and $v : \mathbb{R}^d \times \mathbb{R}_+ \rightarrow \mathbb{R}^d$.

- (a) Show that there exists a Lipschitz function $\sigma : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$ such that $\sigma(x) = P_x^\perp$ for all $\|x\| \geq 1$.

- (b) Consider the natural attempt at spherical Brownian motion given by initializing $X_0 = (1, 0, \dots, 0)$ and setting

$$dX_t = \sigma(X_t)dB_t.$$

for a standard d -dimensional Brownian motion B_t . Show this SDE has a unique solution.

- (c) Use Ito's formula to give an integral expression for $\|X_t\|^2$ and conclude that $\|X_t\| > 1$ for all $t > 0$, almost surely. (Thus, this attempt did not work.)
 (d) Find a constant b (possibly depending on the dimension) such that

$$dX_t = \sigma(X_t)dB_t - bX_t dt$$

Does stay confined to the unit sphere for all time. (Caution: after finding b such that $d\|X_t\|^2 = 0dt + 0dB_t$ **assuming** $\|X_t\| = 1$, you still need to show that the actual solution X_t remains on the sphere. For this, you might find an ODE solved by $\|X_t\|^2$ and argue uniqueness of solutions.)

4. (**Extra Credit**) In class, we used the Lindeberg method to obtain central limit theorems for independent sums and martingales. Here you will investigate further and obtain quantitative bounds.

Let $\varphi : \mathbb{R} \rightarrow [0, 1]$ be a smooth function with $\varphi(x) = 0$ for all $|x| \geq 1$ and $\int_{-1}^1 \varphi(x)dx = 1$. (You may assume such a “bump function” exists, and are encouraged to look them up if unfamiliar.)

Let $\varphi_\epsilon(x) = \varphi(x/\epsilon)/\epsilon$, so that

$$\int_{-\infty}^{\infty} \varphi_\epsilon(x)dx = 1$$

for all ϵ . Note that for each $j \geq 0$ there is $C_j < \infty$ such that the j -th derivative of φ_ϵ is at most

$$\sup_{x \in \mathbb{R}} |\varphi_\epsilon^{(j)}(x)| \leq C_j \epsilon^{-1-j}.$$

- (a) Consider $\epsilon = n^{-a}$ for small $a \in (0, 1)$ (you can choose!) and let $f_{\epsilon,u}$ be the convolution of φ_ϵ with $1_{[u,\infty)}$, i.e.

$$f_{\epsilon,u}(x) = \int_u^\infty \varphi_\epsilon(x-y)dy.$$

Consider an independent sum $S_n = X_1 + \dots + X_n$ with $\mathbb{E}[X_i] = 0$ and $\mathbb{E}[X_i^2] = 1$ for each i , and $\mathbb{E}[|X_i|^3] \leq C$ for all i . Using the Lindeberg method from class, show a convergence rate

$$\sup_u |\mathbb{E}[f_{\epsilon,u}(S_n/\sqrt{n}) - f_{\epsilon,u}(z)]| \leq O(n^{-b})$$

for $b > 0$, where z is a standard Gaussian.

- (b) Deduce uniform convergence (i.e. convergence in L^∞) of the cumulative distribution functions, again at a rate n^{-a} for some $a > 0$. (**Hint** given on last page.)
 (c) Deduce an improved convergence rate for symmetric X_i with uniformly bounded 4-th moment, using matching of the 3-rd moments.

Hints

- For 1: fix small η and apply the martingale CLT from class to a stopped martingale with slightly rescaled last increment, which has total variance exactly $1 - \eta$ whenever $V \geq 1 - \eta$ holds for the original martingale. Then use the L^2 maximal inequality to show this is a sufficiently good approximation.
- For 2(c): to show it is not a true martingale, write an integral expression for $\mathbb{E}[X_t]$ and show it tends to 0 as $t \rightarrow \infty$.
- For 4(b): you may want to use the boundedness of the Gaussian density here.